

CHENGLONG YU

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EDUCATION

Institute of Computing Technology, Chinese Academy of Sciences (ICT, CAS) Beijing, China
Master of Engineering in Computer Technology Sep. 2024 – Expected Jun. 2027
China University of Petroleum (East China) Qingdao, China
Bachelor of Engineering in Computer Science and Technology Sep. 2020 – Jun. 2024
GPA: 4.12/5 | Rank: 3/114 (Top 3%)

RESEARCH INTEREST

Efficient Multimodal Learning, Data-Efficient Vision/World Models, Multimodal Fusion, Knowledge Distillation, Embodied/Edge Intelligence

RESEARCH EXPERIENCE

Efficient World Model-Guided Multi-Agent Decision-Making Dec. 2023 - Nov. 2025

- **Developed** a lightweight multi-agent world model to learn environment dynamics and reward prediction from historical observation-action trajectories.
- **Formulated** multi-agent interaction as a sequence prediction problem and used imagined rollouts to guide policy learning and action planning.
- **Designed** resource-aware action constraints to reduce the policy search space for edge agents under energy and latency limitations.
- **Improved** convergence speed by 36% over MARL baselines and achieved higher energy-efficient policies in multi-task edge scenarios.

Data-Free Knowledge Distillation with Diffusion-Augmented Synthetic Data Aug. 2025 - Present

- **Studied** data-free knowledge distillation for deploying compact neural networks without access to original training data.
- **Proposed** an attention-constrained distillation framework that aligns the student model's focus with teacher-identified semantic regions.
- **Enhanced** synthetic training data with GLIDE-generated images to compensate for the weak semantics of GAN-generated samples.
- **Achieved** state-of-the-art accuracy on CIFAR-10 and CIFAR-100 under data-free distillation settings.

PUBLICATIONS & PATENTS

- C.Yu, W. Xing, J. Shi, Y. Zhou and L. Liu, "Collaborative Multi-Agent Deep Reinforcement Learning for Joint Task Offloading and Resource Allocation with Long Term Energy Control," 2025 IEEE 102nd Vehicular Technology Conference (VTC2025-Fall), Chengdu, China, 2025, pp. 1-5. **(First Author)**
- C.Yu, Y. Zhou, Y. Qi and J. Wang, "Joint Task Offloading and Resource Allocation with Multi-Task Energy Control based on World Model Guided Collaborative Multi-Agent Deep Reinforcement Learning," IEEE Internet of Things Journal, under review. **(First Author)**
- Y. Zhou, C. Yu, J. Shi, Y. Qi, W. Xing, and L. Liu, "An Industrial Internet System for Resource Allocation Based on Collaborative Multi-Agents." Chinese National Invention Patent, Patent No.: CN202511002125.2.
- Y. Qi, Y. Wu, Y. Zhou, C. Yu, et al. "Agent-Oriented Multi-Mode Convergence and AI-Native Orchestration for Low-Altitude Intelligent Networks," IEEE Communications Magazine, accepted.

HONORS & AWARDS

- Outstanding Graduate of Shandong Province - 2024
- National Encouragement Scholarship (Three times) - 2021, 2022, 2023
- UCAS Academic Scholarship (Twice) - 2024, 2025
- Honors Bachelor's Degree, China University of Petroleum (East China) - 2024

TECHNICAL SKILLS

- **Programming & Tools:** Python, C, C++, Pytorch, Linux, Git, CUDA